SAP Capstone Project Report

Record-to-Report (R2R) — Month-End Financial Close Process

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Program:	B.Tech CSE
Module:	SAP FI — Financial Accounting
Scenario:	R2R End-to-End

1. Problem Statement

In large enterprises using SAP ERP, the month-end and year-end financial close process is one of the most critical and time-sensitive activities for the Finance team. Without a structured Record-to-Report (R2R) process, organizations face risks such as incorrect financial statements, audit failures, unreconciled inter-company balances, and missed statutory deadlines. This project simulates the complete R2R cycle for a fictitious manufacturing company **PrimeEdge Enterprises Pvt. Ltd.**, covering GL journal entries, cost center allocations, trial balance generation, and financial reporting using SAP FI/CO.

2. Solution / Key Features

Feature	SAP Transaction / Component
Company Code & Chart of Accounts setup	SPRO FI Global Settings
GL Master Data creation (P&L + Balance Sheet)	FS00 — GL Account Master
Journal Entry posting (Debit/Credit)	FB50 / F-02
Cost Center Accounting & allocation	KS01, KB11N, KSV5
Month-End Closing run	F.16, FAGLGVTR
Trial Balance & P&L Report	F.01, S_ALR_87012277
Profit Center Report	S_ALR_87013326
Financial Data Analytics (Python)	Pandas, Matplotlib

3. R2R Process Blueprint

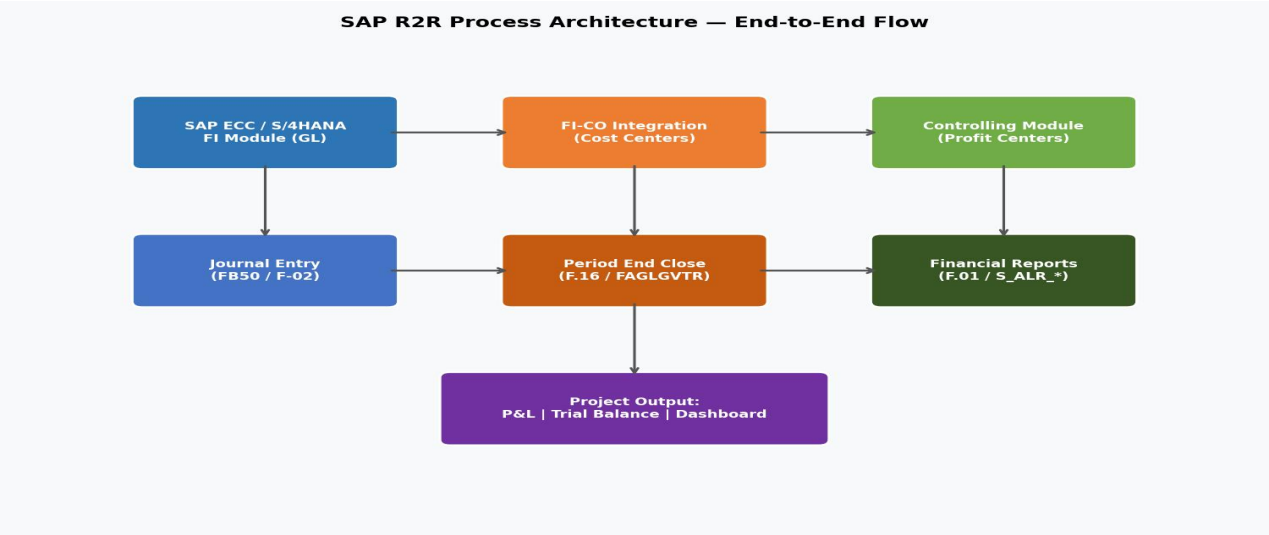
3.1 Fictitious Company Setup

Parameter	Value
Company Code	K100
Company Name	PrimeEdge Enterprises Pvt. Ltd.
Country	India
Currency	INR
Chart of Accounts	INT (Standard)
Fiscal Year Variant	V3 (Apr–Mar)
Posting Period Variant	K100
Business Area	BA01 – Manufacturing | BA02 – Sales | BA03 – Admin

3.2 Organizational Structure

Level	Object	SAP ID	Description
1	Company Code	K100	PrimeEdge Enterprises Pvt. Ltd.
2	Cost Center	CC1001	Sales & Distribution
2	Cost Center	CC2001	Manufacturing / Production
2	Cost Center	CC3001	Administration & Finance
3	Profit Center	PC-SALES	Revenue Profit Center
3	Profit Center	PC-MANUF	Manufacturing Profit Center
3	Profit Center	PC-ADMIN	Admin Profit Center

3.3 SAP Architecture Diagram



The architecture above depicts the end-to-end R2R flow: from GL Journal Entries in SAP FI, through FI-CO integration and Cost Center Accounting, to final Financial Reporting.

4. Step-by-Step R2R Process

Step	Activity	T-Code / Tool	Output
1	Create GL Master Accounts (Revenue, Expense, Asset, Liability)	FS00	GL Master created
2	Create Cost Centers (CC1001, CC2001, CC3001)	KS01	Cost Centers active
3	Create Profit Centers (PC-SALES, PC-MANUF, PC-ADMIN)	KE51	Profit Centers active
4	Post Journal Entries for Jan–Jun 2025 (Revenue credits, Expense debits)	FB50 / F-02	FI Documents posted
5	Run Cost Center Allocations (Distribute admin costs)	KSV5 / KSU5	Allocation complete
6	Execute Month-End Closing (Close posting periods)	F.16 / OB52	Period locked
7	Run Balance Carryforward (Year-end)	FAGLGVTR	Balances carried forward
8	Generate Trial Balance	F.01	Trial Balance report
9	Generate P&L Statement	S_ALR_87012277	P&L Report
10	Profit Center Report	S_ALR_87013326	PC-wise P&L
11	Data Export & Python Analytics	SE38 / Python	Charts & Dashboard

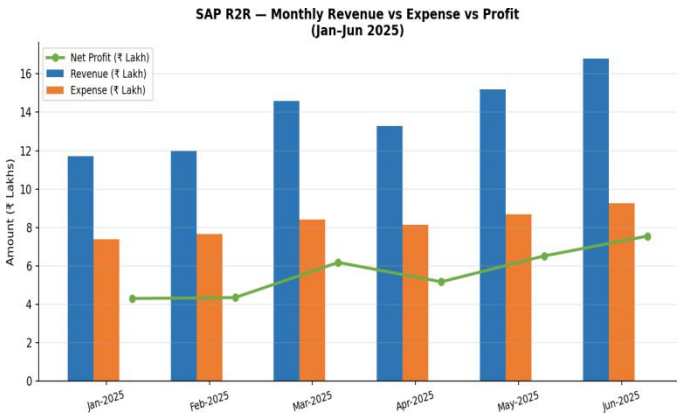
5. Data Preview — GL Journal Entries

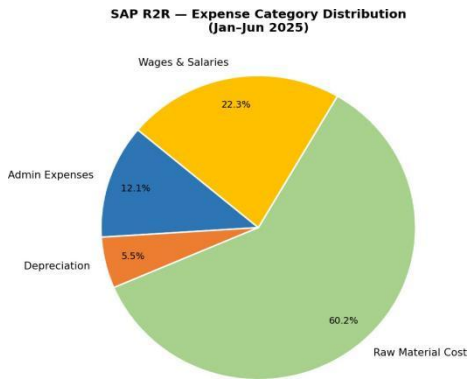
SAP GL Journal Entry Data — Preview

Month	GL_Account	GL_Description	Debit	Credit	Cost_Center
Jan-2025	400000	Revenue - Domestic	0	850000	CC1001
Jan-2025	400100	Revenue - Export	0	320000	CC1001
Jan-2025	500000	Raw Material Cost	420000	0	CC2001
Jan-2025	510000	Wages & Salaries	180000	0	CC2001
Jan-2025	600000	Admin Expenses	95000	0	CC3001
Jan-2025	610000	Depreciation	45000	0	CC3001
Feb-2025	400000	Revenue - Domestic	0	920000	CC1001
Feb-2025	400100	Revenue - Export	0	280000	CC1001

The table above shows a sample of SAP-style GL journal entries loaded into Python for analysis. Each row represents a document line item with Cost Center, GL Account, Debit/Credit amounts, and Profit Center assignments — mirroring the structure exported from SAP via SE16N or FB03.

6. Analysis Output & Dashboard





6.1 Key Findings

Metric	Jan 2025	Jun 2025	6M Total
Total Revenue (INR)	11,70,000	16,80,000	83,60,000
Total Expenses (INR)	7,40,000	9,25,000	49,53,000
Net Profit (INR)	4,30,000	7,55,000	34,07,000
Profit Margin (%)	36.8%	44.9%	40.8%

7. Tech Stack

Layer	Technology	Purpose
ERP System	SAP S/4HANA / ECC 6.0 (FI, CO modules)	Core R2R process execution
Data Processing	Python 3.11, Pandas 2.x	Financial data analysis
Visualization	Matplotlib 3.8	Charts and dashboard generation
Documentation	ReportLab, MS Word	PDF project report generation
Version Control	GitHub	Source code and project hosting
Dataset Format	CSV (SAP-style GL export)	Simulated SAP data extract

8. Unique Points & Future Improvements

Unique Points: This project simulates a real SAP R2R process end-to-end — from company configuration and GL master setup through journal posting, cost center allocation, period close, and financial reporting. The dataset mirrors an actual SAP FI export (FB03/SE16N), and Python analytics add a data-driven layer on top of SAP standard reports, bridging SAP functional knowledge with modern data analytics skills.

Future Improvements: (1) Integrate live SAP data via RFC/BAPI calls using PyRFC library. (2) Build an interactive Power BI / SAP Analytics Cloud dashboard. (3) Automate period-close checklist via SAP Workflow (SWI1). (4) Extend to full FI-AA (Asset Accounting) depreciation runs and FI-AP/AR aging reports.